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Task Description:	Clear mechanistic understanding of the biological processes elicited by high LET radiation increase cancer risk will reduce the uncertainties in assessing the risk of space flight. Classic targeted radiation effects like cell kill and mutation are proportional to dose and their LET dependence is described by relative biological effectiveness. However recent data suggests that high LET radiation affects multicellular processes that promote cancer. We propose that radiation carcinogenesis results from two classes of ionizing radiation effect: targeted effects that alter genomic sequence and non-targeted effects (NTE) that alter phenotype and multicellular interactions. Since NTE are not proportional to dose, new approaches are required to assess the impact of NTE, which alter gene expression, cell signaling, genomic stability and cell phenotypes, on carcinogenesis. It is unknown to what extent NTE contribute to the significantly greater carcinogenic effect of high LET radiation. We hypothesize that NTE promotes epithelial cancer. Project 1 will test use novel transgenic and humanized mouse mammary models to evaluate the relative contribution of radiation targeted and NTE to carcinogenesis as a function of radiation quality and of age a radiation chimera model. We will monitor inflammation as a key process affected by radiation NTE and conduct a systems biology analysis of mammary gland. Project 2 will employ new systems genetic analysis of cancer susceptibility in radiation sensitive organs to define critical regulatory networks that impact the relative carcinogenic effectiveness of high LET radiation. Project 3 will use Monte Carlo simulations of targeted radiation effects and agent based modeling of cell interactions to model the relative contributions of targeted and NTE to carcinogenesis. This NSCOR will provide important data on radiation quality dependence of targeted and non-targeted radiation effects and the genetic determinants of cancer susceptibility, which together will used to reduce uncertainties in NASA radiation risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	First, radiation can be considered an experimental probe that perturbs tissue function. Studying these perturbations is a means to define critical biological response to stress. Second, we will develop a comprehensive data base of normal tissue and irradiated tissue by analyzing different organs in each mouse. This data base will become a public resource. Third, our studies will identify genetic predisposition to radiation associated cancer, and may thus help protect humans who have the greatest risk of cancer.
	Introduction This NSCOR provides experimental and modeling studies to define the efficiency and physiological context in which high LET radiation increases epithelial cancer risk. We hypothesize that both targeted and NTE contribute to radiation carcinogenesis. Targeted effects of ionizing radiation result from interaction of energy with DNA, leading to DNA damage responses and changes in genomic sequence. On the other hand, poorly understood NTE alter phenotype and multicellular interactions that could contribute to cancer. We hypothesize that the greater carcinogenic risk from high LET irradiation is due to NTE promotion of targeted radiation effects. The data generated the NYU/LBNL/UCSF NSCOR will provide a comprehensive analysis of LET dependence for solid tumor frequency in different tissues, integrated with computational modeling and systems biology to identify key events. Project 1 Progress We used radiation/genetic mammary chimera mouse model to separate targeted effects in the organ from the non-targeted effects on host cells. This model consists of unirradiated Tp53 null mammary epithelium transplanted into the inguinal cleared mammary fat pad of wild type host mice. Host mice were irradiated three days before transplantation with one of the following: 100 cGy gamma-rays, or 29 or 81cGy 600MeV/amu Fe corresponding to a fluence of approximately 1 and 3, particles respectively per cell, or 10, 29 or 81cGy 350MeV/amu Si corresponding to a fluence of approximately 1, 3 and 7 particles respectively per cell at the NSRL. Controls were sham-irradiated at NSRL. These mice were monitored for 600 days for subsequent development of palpable tumors. Upon detection of a first mammary gland tumor, a survival surgery was performed allowing the opportunity for a second tumor to develop on the contralateral gland. At experiment termination, we determined which transplants failed, which were eliminated from analysis. To evaluate the contribution of the radiation effects on epithelium (so-called targeted effects like mutation), 204 fat-pads were cleared and transplanted with Trp53 null epithelium at 3 weeks of age were irradiated at 10 weeks of age at NSRL10-C with 30 cGy and 82 cGy 600Mev/amu Fe at or 10, 50, 100 cGy gamma-radiation. To evaluate the effect of age in the radiation chimera-experiment, mice were irradiated at 10 months, simulating the average age for an astronauts. For this purpose 200 mice were cleared at 3 weeks of age and then housed under standard conditions. At 10 months the mice were irradiated at NSRL 11B with three different ions (81 cGy Fe 600Mev/amu, 49 cGy Ar and 29 cGy Si 350Mev/amu) or three dose doses of gamma-radiation (10, 50, 100 cGy). Three days after irradiation the inguinal glands were transplanted with Trp53 null fragments. At this point the mice are being palpated for tumor detection and we expect to complete this experiment by fall 2012. Project 2 Progress To study the long-term impact of high LET compared to low LET radiation upon tumor formation we began a long-term tumorigenesis study in November 2010 using p53 deltaP mice, aged between 6-10 weeks, upon a mixed background (129/Sv/C57BL/6). 110 p53 deltaP mice were irradiated with either 29, 81, or 100 cGy 600 MeV/amu Fe-ion particles or 100 cGy gamma-radiation. In addition, wild-type controls of 129/Sv or C57BL/6 mice were irradiated. At the time of sacrifice radiosensitive tissues including the skin, mammary, thymus and spleen were harvested along with the arising tumors, which will be used for gene expression analysis, sequencing and histology. Furthermore, blood samples are taken for blood smears, complete blood count and pellets and serum. Our results show that over time there is no statistically significant difference between the 29, 81, 100 cGy Fe-particle exposure and 100 cGy gamma-radiation exposure upon tumor-free survival. The most prevalent tumor types seen were sarcomas, carcinomas and lymphomas. These have been seen in various organs including the skin, mammary gland, lung, liver, spleen, thymus and thyroid. At this point we have not observed a different pattern of tumor type or tissue of origin induced by different doses of Fe-ion or gamma-radiation. RNA and DNA are currently being extracted from these tumor samples for gene expression and sequencing analysis. We have previously identified a unique pattern of genomic alterations arising in low LET radiation-induced lymphomas in Trp53 null mice. In order to enable direct comparisons with our previous data we have successfully bred the p53

Task Progress:	deltaP mice onto a 99.9% pure FVB/N background. Currently, there are no studies describing the impact of high LET radiation upon the FVB genetic background, therefore these studies will provide valuable information as to the impact of genetic background upon heavy-ion radiation-induced carcinogenesis. In June and November 2011 we irradiated 250 FVB/N mice of genotypes p53 deltaP, and Trp53 wildtype and heterozygote mice with 50 cGy of either 600 MeV/amu Fe-ion particles or gamma radiation. These studies are still ongoing, and a range of tumors in radiosensitive tissues including the skin, liver, lung, mammary, thymus, thyroid and spleen has been observed in the p53 deltaP and Trp53 heterozygote mice. Currently we are extracting DNA and RNA from these tumors for gene expression and genomic analysis. We completed a short-term time course in April 2012 using 125 mice. Consistent with previous experiments we used a dose of 50 cGy and irradiated with both 600 MeV/amu Fe-ions and gamma ions. Four genotypes of FVB/N mice were irradiated: p53 deltaP, Trp53 heterozygote, null and wildtype. These animals were then sacrificed at times 6, 24, 48 hours and 1 week post-radiation, along with an un-irradiated control set. At the time of necropsy a blood sample was taken and four tissues were harvested (skin, mammary, thymus and tail). Sections of these tissues were both fixed for histological analysis and frozen. RNA will be extracted from frozen tissues and gene expression levels at various time points post-radiation in the different genotypes will be investigated. This is an exploratory time-point experiment which will guide the time post-radiation at which we sacrifice our backcross population animals, for gene expression network analysis. We proposed to perform a systems genetic analysis of susceptibility to tumors induced by high LET using mice from a backcross set up between Mus spretus and FVB/N p53 deltaP. Unfortunately, we have encountered unexpected practical difficulties with generating this p53 deltaP F1 backcross progeny. Breeding was problematic, with very few litters being born, and small numbers of pups within each litter. Furthermore, upon genotyping it became apparent that there was a significant level of lethality associated with the homozygous p53 deltaP backcross animals. This was surprising, as we have been able to breed the Trp53 null allele in a similar cross without seeing any evidence for lethality in this background. As a result, we were unable to breed sufficient numbers of F1bx Mus spretus/FVB/N p53 deltaP as would be required to have significant statistical power for analysis. Hence we have changed the Spretus backcross studies to analysis of the Trp53 heterozygous mice. These data will also directly complement the studies in Project 1 which use a Trp53 null mouse model. Currently we are initiating a backcross between Mus spretus and FVB/N Trp53 heterozygotes, which we will expose to 50 cGy 600 MeV/amu Fe-ions and ion-radiation. From the tissue samples taken pre- and post-radiation we will extract RNA and investigate the perturbations in tissue gene expression networks after exposure to high and low LET radiation. This will allow us to generate a unique view of gene expression networks and enable us to identify components that impact upon sensitivity to tumor development post-exposure to both low and high LET radiation. Ultimately, these data, together with the genomic analysis of tumors induced by both high and low LET radiation described above, will assist in the identification of potential susceptibility genes and pathways for future assessment in human populations. Project 3 Progress As described in Nguyen et al., a single low dose of radiation increases mammary stem cell frequency in mice. It is not clear how processes induced by radiation affect stem cell regulation. In order to evaluate the long-term impact of radiation-induced death/senescence and stem cell self-renewal signaling in the mammary gland, so we integrated agent based modeling (ABM) with in vivo measurements and in vitro cell culture data. In silico modeling of mammary morphogenesis using ABM allows events to be evaluated in the context of development, with its incumbent complexity of lineage heterogeneity, proliferation and morphogenesis. Expression profile data from Project 1 were analyzed from mammary glands of mice irradiated 1, 4 and 12 weeks before by sparsely ionizing gamma-ray radiation (1 Gy) or densely ionizing 350 MeV/amu Si particle or 600 MeV/amu Fe particle. We identified enrichment for a MaSC signature at 1 week post-IR regardless of radiation quality. At 4 weeks post-IR, the number of MaSC genes decreased. At 12 weeks, MaSC gene enrichment was no longer evident in any group. These analyses are ongoing in collaboration with Project 1. We benchmarked the ABM using mammary gland measurements (e.g. branching frequency, time of development). Model parameters (e.g. cell doubling time, differentiation probabilities) were determined by generating simulations matching in vivo measurements from tissues representing maturation from puberty to adult (weeks 3 to 9). We then tested various mechanisms thought to be involved in the radiation-induced stem cell enrichment. ABM rejected cell loss via death or replicative senescence as a source for stem cells increase, and supported a small, transient increase in TGFß and Notch activity as an effective means to stimulate self-renewal. Preliminary analysis of expression profiles of tumors from Project 1 suggests that many biological processes were affected by HZE host irradiation. Pathway analysis of the genes associated with tumors from Si-irradiated host invoked cell growth and proliferation, hematological disease, hematological system development, cellular assembly and organization, and metabolic disease. These tumors were also significantly enriched for a mammary stem cell signature. These processes and signatures are similar to those we previously identified in tumors arising in gamma-irradiated hosts (Nguyen et al. 2011. Radiation Acts on the Microenvironment to Affect Breast Carcinogenesis by Distinct Mechanisms that Decrease Breast Cancer Latency and Affect Tumor Type. Cancer cell, In press). Bioinformatics Components We switched from Affymetrix Mouse 430 2.0 arrays to Affymetrix Mouse gene 1.0 ST arrays at the UCSF Cancer Center. The latter is about half the cost, due to its ability to process more samples simultaneously. This newer technology is also fully compatible with the older arrays and in fact brings new elements that will benefit this proposal. However, there are fewer analytic tools for this platform so we have begun to generate our own analytical tools for the information from Mouse Gene 1.0 ST Arrays.
Bibliography Type:	Description: (Last Updated: 11/01/2023)

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